Homework 4. Moving Average Process

You can get up to 10 points for this homework.

Question 1. For the following tasks use Python and submit one file incorporating all your answers, named as "H4-YourFirstName-YourLastName.ipynb".

- [A] Simulate a stationary MA(3) process $y_t = 0.9\theta_{t-1} + 0.3\theta_{t-2} + 0.2\theta_{t-3}$. To do so, use the ArmaProcess function from the statsmodels library.
- [B] Generate 10,000 samples.
- [C] Plot your simulated moving average.
- [D] Run the ADF test, and check if the process is stationary.
- [E] Plot the ACF, and see if there are significant coefficients after lag 3
- [F] Separate your simulated series into train and test sets. Take the first 9,800 timesteps for the train set, and assign the rest to the test set.
- [G] Make forecasts over the test set. Use two baseline models of your choice, and an MA(3) model. Make sure you repeatedly forecast 3 timesteps at a time.
- [H] Plot your forecasts on the same graph.
- [I] Measure the MSE, identify your champion model and draw the MSEs in a bar plot.